

MATH 2790 Fall 2025

Homework 13

Due date: Wednesday, Dec 10

Instructions:

- (a) It is expected to show detailed work for every problem. You are allowed to have a discussion with peers on some challenging questions. But then you should work on your own to finish the assignment.
- (b) All questions in this assignment are from sections 8.3 and 9.2.

Problems:

- 1. Question (2) on page 435 of textbook under section 8.3.
- 2. Question (3) on page 435 of textbook under section 8.3.
- 3. Question (5) on page 435 of textbook under section 8.3.
- 4. Question (2) on page 496 of textbook under section 9.2.
- 5. Question (3) on page 496 of textbook under section 9.2.
- 6. Question (4) on page 496 of textbook under section 9.2. It should be a par-value bond.
- 7. On December 31, year X , your company estimates that it will pay a total of \$10 million in year $X + 1$ and subsequent years (from year $X + 2$ to year $X + 4$) on health insurance claims that were incurred in year X . The payout pattern is as follows:

Year	$X + 1$	$X + 2$	$X + 3$	$X + 4$
Payout percentage	60%	20%	10%	10%

Determine the Macaulay duration of the claim payments assuming that they are paid at the end of each calendar year and the effective rate of interest is 10%.